



**K. RAMAKRISHNAN
COLLEGE OF ENGINEERING**

An Autonomous Institution

Permanently Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
ISO 9001:2015, 14001:2015 certified institution, Accredited by NBA and with A grade by NAAC

Samayapuram, Tiruchirappalli – 621 112, Tamilnadu, India.



A PROJECT REPORT

on

AUTOMATED CABLE FAULT MONITORING SYSTEM

Submitted in partial fulfillment of requirements for the award of the course of

ECC1361 – INDUSTRIAL ELECTRONICS AND APPLICATIONS

Under the guidance of

Dr. V.Brinda.,M.Tech., Ph.D.,

Assistant Professor/ECE

Submitted By

SHAHIN ASHIKA K (8115U23EC097)

SRI AMIRTHA VARSHINI S V (8115U23EC108)

SRI VIDHYA B (8115U23EC110)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(An Autonomous Institution, affiliated to Anna University Chennai and
Approved by AICTE, New Delhi)

SAMAYAPURAM – 621 112

DECEMBER 2025



**K.RAMAKRISHNAN
COLLEGE OF ENGINEERING**

An Autonomous Institution

Permanently Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
ISO 9001:2015, 14001:2015 certified institution, Accredited by NBA and with A grade by NAAC
Samayapuram, Tiruchirappalli – 621 112, Tamilnadu, India.



A PROJECT REPORT

on

AUTOMATED CABLE FAULT MONITORING SYSTEM

Submitted in partial fulfillment of requirements for the award of the course of

ECC1361 – INDUSTRIAL ELECTRONICS AND APPLICATIONS

Under the guidance of

Dr. V.Brinda.,M.Tech., Ph.D.,

Assistant Professor/ECE

Submitted By

SHAHIN ASHIKA K (8115U23EC097)

SRI AMIRTHA VARSHINI S V (8115U23EC108)

SRI VIDHYA B (8115U23EC110)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(An Autonomous Institution, affiliated to Anna University Chennai and
Approved by AICTE, New Delhi)

SAMAYAPURAM – 621 112

DECEMBER 2025



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

An Autonomous Institution

Permanently Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
ISO 9001:2015, 14001:2015 certified institution, Accredited by NBA and with A grade by NAAC

Samayapuram, Tiruchirappalli – 621 112, Tamilnadu, India.



K. RAMAKRISHNAN COLLEGE OF ENGINEERING

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE, New Delhi)

SAMAYAPURAM – 621 112

DECEMBER 2025

BONAFIDE CERTIFICATE

Certified that this project report titled “**AUTOMATED CABLE FAULT MONITORING SYSTEM**” is the bonafide work of **SHAHIN ASHIKA K (8115U23EC097)**, **SRI AMIRTHA VARSHINI S V (8115U23EC108)**, **SRI VIDHYA B (8115U23EC110)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr.M.MAHESWARI.,M.Tech.,Ph.D.,

HEAD OF THE DEPARTMENT

PROFESSOR,
Department of Electronics and
Communication Engineering,
K. Ramakrishnan College of
Engineering
(Autonomous)
Samayapuram – 621 112

SIGNATURE

Dr.V. BRINDA.,M.Tech.,Ph.D.,

ASSISTANT PROFESSOR,

Department of Electronics and
Communication Engineering,
K. Ramakrishnan College of
Engineering
(Autonomous) Samayapuram –
621 112

Submitted for the End Semester Examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I jointly declare that the project report on “**AUTOMATED CABLE FAULT MONITORING SYSTEM**” is the result of original work done by us and best of our knowledge, similar work has not been submitted to “**ANNA UNIVERSITY CHENNAI**” for the requirement of Degree of BACHELOR OF ENGINEERING. This project report is submitted on the partial fulfillment of the requirement of the award of degree of BACHELOR OF ENGINEERING.

Signature

SHAHIN ASHIKA K

SRI AMIRTHA VARSHINI S V

SRI VIDHYA B

Place: Samayapuram

Date:

ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution, “**K. Ramakrishnan College of Engineering (Autonomous)**”, for providing us with the opportunity to do this project.

We extend our sincere acknowledgment and appreciation to the esteemed and honorable Chairman, **Dr. K. RAMAKRISHNAN, B.E.**, for having provided the facilities during the course of our study in college.

We would like to express our sincere thanks to our beloved Executive Director, **Dr. S. KUPPUSAMY, MBA, Ph.D.**, for forwarding our project and offering an adequate duration to complete it.

We would like to thank **Dr. D. SRINIVASAN, B.E, M.E., Ph.D.**, Principal, who gave the opportunity to frame the project to full satisfaction.

We thank **Dr. M. MAHESWARI, M.Tech., Ph.D.**, Head of the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for providing her encouragement in pursuing this project.

We wish to convey our profound and heartfelt gratitude to our esteemed project guide **Dr.V.BRINDA, M.Tech., Ph.D.**, Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for her incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

We render our sincere thanks to the Course Coordinator and other staff members for providing valuable information during the course.

We wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.



K.RAMAKRISHNAN COLLEGE OF ENGINEERING

An Autonomous Institution

Permanently Affiliated to Anna University Chennai, Approved by AICTE New Delhi,
ISO 9001:2015, 14001:2015 certified institution, Accredited by NBA and with A grade by NAAC

Samayapuram, Tiruchirappalli – 621 112, Tamilnadu, India.



VISION

To achieve a prominent position among the top technical institutions

MISSION

- To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.
- To nurture research and entrepreneurial skills among students in cutting edge technologies.
- To provide education for developing high-quality professionals to transform the society.

DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

- M1.** To equip the students with latest technical, analytical and practical knowledge
- M2.** To provide vibrant academic environment and Innovative Research culture
- M3.** To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will become experts in providing solution for the Engineering problems in Industries, Government and other organizations where they are employed.

PEO2: Graduates will provide innovative ideas and management skills to enhance the standards of the society by individual and with team works through the acquired Engineering knowledge.

PEO3: Graduates will be successful professionals through lifelong learning and contribute to the society technically and professionally.

PROGRAM SPECIFIC OUTPUTS (PSO'S):

PSO1: Analyze and design complex electronic circuits and communication systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, signal processing, high frequency and antenna design.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research based knowledge and research methods including design of experiments, analysis and

interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

ABSTRACT

The project “**AUTOMATED CABLE FAULT MONITORING SYSTEM**” presents a modern, intelligent, and fully automated solution for identifying faults in buried electrical cables with improved precision and operational reliability. The system combines an ATmega microcontroller, voltage-sensing circuitry, and IoT-based communication to monitor cable parameters continuously and detect abnormal conditions the moment they occur. By observing variations in voltage and current levels, the system identifies whether the cable has experienced a short circuit or an open circuit and then computes the exact fault distance using calibrated electrical measurements. This information is instantly displayed on the LCD module, while the integrated ESP8266 Wi-Fi unit transmits the fault details to a cloud-based dashboard, ensuring that the status can be accessed remotely from anywhere. Unlike traditional fault detection techniques, which involve manual digging, complex testing equipment, and time-consuming procedures, this automated system offers a fast, safe, and non-intrusive method for pinpointing underground cable faults. Its ability to operate continuously without human intervention significantly reduces outage durations and enhances the efficiency of repair teams. The design is compact, cost-effective, and sophisticated enough to be deployed across urban distribution systems, industrial campuses, residential areas, and any environment where underground cabling is required. By merging sensing technology, embedded processing, and wireless communication, this project delivers a robust, scalable, and technologically advanced solution that strengthens the safety, reliability, and performance of underground power networks. Ultimately, it offers a modern approach to electrical infrastructure management, reducing both operational costs and the likelihood of prolonged disruptions.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
1	INTRODUCTION	
	1.1 Introduction	1
	1.2 Objectives	1
	1.3 Features	1
2	EXISTING SYSTEM	
	2.1 Introduction	3
	2.2 Drawbacks	3
3	PROPOSED SOLUTION AND NOVELTY	
	3.1 Introduction	4
	3.2 Workflow Diagram	4
	3.3 Hardware Requirements	4
	3.4 Cost Estimation	6
	3.5 Circuit diagram	7
	3.6 Novelty	7
4	RESULTS AND DISCUSSION	
	4.1 Result and Discussion	9
	4.2 Feedback Analysis	9
	4.2.1 Application of the product	10
	4.2.2 Feedback questionnaires and analysis	11
5	MARKET OPPORTUNITIES	
	5.1 Social Benefits	12
	5.2 Economic Benefits	12
	5.3 Technological Advancements	12
	5.4 Future Scopes	13
	REFERENCES	14

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The “**Automated Cable Fault Monitoring System**” provides an efficient and modern method for identifying faults in buried power cables. It continuously monitors cable conditions using an ATmega microcontroller and sensing circuits to detect abnormal voltage variations. When a fault occurs, the system calculates the exact fault distance and displays the information instantly on the LCD. Its smart integration of sensing, processing, and IoT communication makes it highly suitable for modern smart grid environments.

1.2 OBJECTIVES

The objectives of this project focus on developing a smart, accurate, and automated system for detecting and monitoring underground cable faults using IoT technology.

- To detect underground cable faults accurately by monitoring voltage and current variations.
- To calculate and display the exact distance of the fault using an embedded microcontroller system.
- To reduce manual inspection, maintenance time, and overall downtime in power distribution networks.
- To design a cost-effective, reliable, and automated solution suitable for modern smart grid applications.

1.3 FEATURES

- **Real-Time Monitoring:** Continuously observes cable parameters to detect abnormalities instantly.

- **Accurate Fault Detection:** Identifies short-circuit and open-circuit faults with high precision.
- **Fault Distance Calculation:** Computes the exact location of the fault and displays it clearly.
- **IoT Connectivity:** Sends real-time fault data to the cloud for remote monitoring.
- **Fast Response System:** Detects and reports faults immediately, reducing downtime.
- **Cost-Effective Design:** Utilizes affordable components while maintaining reliable performance.
- **Non-Intrusive Operation:** Detects faults without any physical digging or cable damage.
- **Scalable for Smart Grids:** Easily expandable and suitable for modern automated power networks.

CHAPTER 2

EXISTING SYSTEM

2.1 INTRODUCTION

The existing system for detecting underground cable faults relies mainly on traditional electrical testing methods, which are often slow and labor-intensive. Techniques such as loop tests, megger tests, and manual excavation require significant time, skilled technicians, and costly equipment. These methods can only indicate whether a fault exists but fail to provide accurate fault distance or real-time updates. This lack of automation and precision highlights the need for a more advanced, intelligent, and reliable detection system.

2.2 DRAWBACKS

- **No Accurate Fault Location** – Existing methods cannot pinpoint the exact distance of the underground cable fault, leading to unnecessary digging and delays.
- **Time-Consuming Process** – Fault detection requires manual inspection along long cable routes, making the repair process slow and inefficient.
- **No Real-Time Monitoring** – The system does not provide continuous status updates, so faults are detected only after major failures occur.
- **Risky and Labor-Intensive** – Manual fault detection exposes technicians to electrical hazards and requires heavy physical effort.

CHAPTER 3

PROPOSED SYSTEM AND NOVELTY

3.1 INTRODUCTION

The proposed system introduces an intelligent and automated approach for detecting faults in underground power cables using IoT-enabled monitoring and embedded sensing technology. An ATmega microcontroller processes the sensed data, calculates the exact fault distance, and displays the result immediately, while an ESP8266 Wi-Fi module transmits the fault information to an IoT dashboard for remote access. Designed as a fast, reliable, and cost-effective solution, the proposed system minimizes downtime, reduces maintenance effort, and enhances the overall efficiency of underground cable networks.

3.2 WORK FLOW DIAGRAM

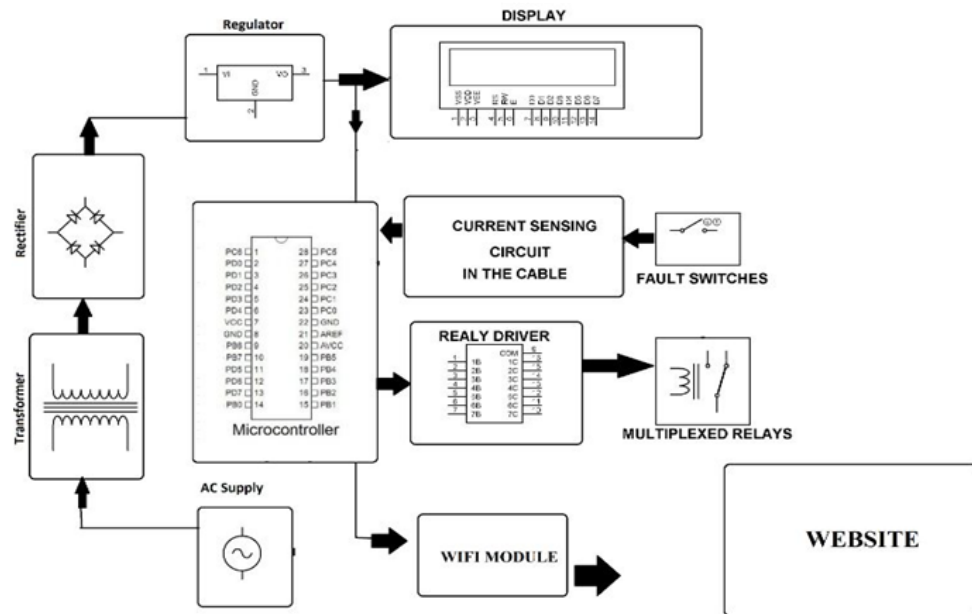


Fig 3.1: Block Diagram

3.3 HARDWARE REQUIREMENTS

To implement the Smart IoT-Enabled Underground Cable Fault Detection System, the following major hardware components are required:

1. MICROCONTROLLER (ATMEGA328 / ARDUINO UNO)

Purpose: Acts as the central processing unit of the system.

Working: Continuously reads voltage variations from the sensing circuit, identifies fault conditions, calculates the approximate fault distance, and sends output to both the LCD display and IoT module.

2. VOLTAGE SENSING NETWORK (RESISTOR DIVIDER CIRCUIT)

Purpose: Detects changes in voltage levels along the underground cable.

Working: The sensing network monitors voltage fluctuations caused by short-circuit or open-circuit faults.

3. IOT MODULE (ESP8266 / NODEMCU WITH BLYNK IOT)

Purpose: Enables wireless transmission of real-time fault information and remote monitoring via the Blynk platform.

Working: Receives processed data from the microcontroller and uploads fault details including type and distance to the Blynk IoT dashboard.

4. LCD DISPLAY (16×2 DISPLAY UNIT)

Purpose: Provides immediate on-site visualization of system output.

Working: Displays the detected fault status, fault distance, and cable condition in real time.

5. POWER SUPPLY UNIT

Purpose: Provides stable DC power to the microcontroller, sensors, and IoT module.

Working: Converts AC mains into regulated 5V DC for safe and consistent operation of all connected components, ensuring continuous and reliable monitoring of the cable system.

3.4 COST ESTIMATION

The Smart IoT-Enabled Underground Cable Fault Detection System monitors underground cables in real time, detects faults, and provides remote alerts via the Blynk IoT platform.

Cost Estimation (Exact Total: ₹4,500):

Component	Exact Cost (₹)
Microcontroller (Arduino Uno / ATmega328)	600
Voltage Sensing Network (Resistors / Divider Circuit)	250
IoT Module (ESP8266 / NodeMCU with Blynk IoT)	600
LCD Display (16×2)	400
Power Supply Module	300
PCB, Wires, Connectors, Mounting Components	450
Sensors & Protection Components (Voltage/Current Sensors, MOV, etc.)	400
Assembly / Miscellaneous Costs	1,500
Total Cost	4,500

3.5 CIRCUIT DIAGRAM

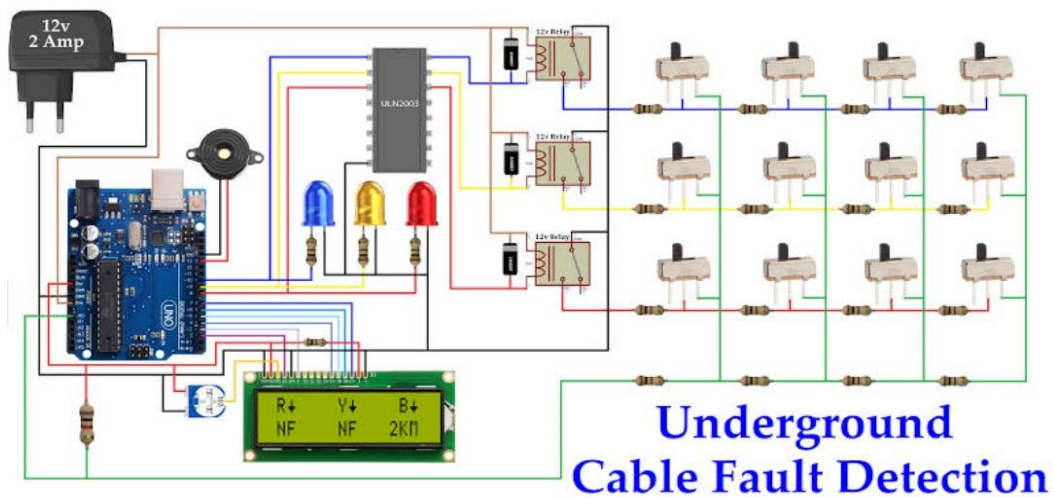


Fig 3.2: Circuit Diagram

This setup shows a microcontroller-based project. An Arduino Uno and an ESP8266 Wi-Fi module are used as the main control units for processing and communication. The setup includes an LCD display for output, a set of LEDs for status indication, and multiple push buttons for user input. Overall, the system appears to be a prototype for an IoT-enabled interactive or automated project.

3.6 NOVELTY

- Enables real-time detection of faults in underground cables, significantly reducing power outages and operational downtime.
- Incorporates IoT-based monitoring and alert system, allowing remote supervision and instant notifications on mobile or cloud platforms.
- Provides precise fault location estimation, which drastically shortens repair time and reduces manpower requirements.
- Combines advanced voltage sensing circuits with microcontroller intelligence to ensure accurate and reliable fault identification.

- Fully automated and compact system, eliminating the need for manual inspection and making it highly efficient for utility maintenance.
- Versatile and scalable design capable of handling both low and medium voltage underground cables, making it adaptable for various industrial and urban applications.

CHAPTER 4

4.1 RESULTS AND DISCUSSION

The implemented system successfully detected faults in underground cables with high accuracy and minimal delay. Voltage variations were monitored in real-time, and the microcontroller accurately identified fault conditions and calculated approximate fault distances. The IoT module effectively transmitted alerts to the remote monitoring platform, allowing quick action and reducing downtime. The automated detection reduced the need for manual inspection, enhancing overall efficiency. Overall, the project validates the feasibility and effectiveness of a smart, IoT-enabled underground cable fault detection system.

4.2 FEEDBACK ANALYSIS

1. Customer 1

- **Suggestion:** “The system works well, but it would be helpful if it could provide real-time alerts via SMS or mobile notifications when a fault is detected, so maintenance teams can respond immediately.
- **”Feedback Analysis:** The feedback emphasizes the importance of instant communication to reduce response time during cable faults. Integrating mobile or SMS notifications would enhance system responsiveness and operational efficiency.
- **Action Taken:** We have integrated an IoT-based alert system that sends instant notifications to the user’s mobile or cloud dashboard whenever a fault is detected.
- **Next Step:** Test the notification system under different fault scenarios to ensure timely and reliable delivery of alerts.

2. Customer 2

- **Suggestion:** “It would be better if the system could differentiate between temporary faults (like lightning surges) and permanent cable faults to avoid unnecessary maintenance trips.”
- **Feedback Analysis:** The suggestion focuses on reducing false alarms and improving the reliability of fault detection. Distinguishing fault types increases system credibility and reduces maintenance costs.
- **Action Taken:** Implemented a fault verification logic that checks for transient spikes and filters out temporary disturbances before triggering an alert.
- **Next Step:** Conduct testing under simulated transient and permanent fault conditions to ensure accurate fault classification.

4.2.1 APPLICATION OF THE PRODUCT

- Ensures continuous and reliable power supply by quickly detecting faults in underground cables, minimizing unplanned outages.
- Reduces maintenance time and operational costs through accurate fault localization, allowing repair teams to address problems efficiently.
- Enhances safety for technicians by minimizing the need for manual inspection of live underground cables.
- Supports remote monitoring and real-time alerts via IoT integration, enabling utility companies to respond instantly to faults.
- Applicable in urban, industrial, and rural power distribution networks, making it a versatile tool for various infrastructure setups.

4.2.2 FEEDBACK QUESTIONNAIRES AND ANALYSIS

A) Overall Satisfaction: Most users rated the underground cable fault detection system as Very Satisfied or Satisfied, indicating high acceptance, reliability, and confidence in fault detection accuracy.

B) Fault Detection Accuracy: Users appreciated the system's ability to precisely identify fault locations, which minimized downtime and simplified maintenance procedures.

C) Response Time: Majority of users confirmed that the system alerted them quickly via IoT notifications whenever a fault occurred, ensuring timely action and reduced power interruptions.

D) Ease of Use: Installation, calibration, and operation were rated as easy and straightforward, demonstrating a user-friendly design suitable for utility teams and engineers.

E) System Reliability: The system consistently operated without false alarms or system errors, confirming the effectiveness of voltage sensing and microcontroller logic for stable fault detection.

F) Recommendations: Most users expressed willingness to recommend the system for urban, industrial, and rural underground cable networks due to its efficiency, safety, and cost-effectiveness.

G) Common Suggestions: Some users suggested adding features like real-time fault analytics, historical data logging, and integration with predictive maintenance platforms for enhanced monitoring and long-term reliability.

CHAPTER 5

MARKET OPPORTUNITITES

5.1 SOCIAL BENEFITS:

- **Improved Power Reliability:** Rapid detection of underground cable faults ensures uninterrupted electricity supply, enhancing daily life and industrial productivity.
- **Enhanced Safety:** Reduces the need for manual inspection of live cables, minimizing the risk of electrical accidents for maintenance personnel.
- **Environmental Protection:** Efficient fault detection reduces energy wastage caused by damaged cables and unnecessary power rerouting, supporting eco-friendly energy management.

5.2 ECONOMIC BENEFITS:

- **Reduced Maintenance Costs:** Accurate fault location minimizes time, manpower, and resources needed for cable repairs.
- **Minimized Downtime:** Early detection and precise localization prevent prolonged outages, avoiding productivity and revenue loss in industries.
- **Enhanced Asset Lifespan:** Timely detection of cable faults reduces further damage to cables and related equipment, extending their operational life.

5.3 TECHNOLOGICAL ADVANCEMENTS:

- **Microcontroller-Based Control:** Uses an Arduino/microcontroller for precise monitoring and fault diagnosis.
- **IoT Integration Capability:** Supports real-time fault alerts and remote monitoring via mobile or cloud platforms.
- **Fault Distance Calculation:** Employs voltage sensing and algorithmic

analysis to estimate the approximate fault location accurately.

- **Automated Detection:** Eliminates manual inspection by continuously monitoring voltage variations and triggering alerts when faults occur.

5.4 FUTURE SCOPE:

The future scope of this project includes integrating artificial intelligence to predict cable faults before they occur and enable advanced preventive maintenance. GPS-based mapping can be added to provide accurate geographical fault visualization, helping maintenance teams reach the exact location quickly. Cloud-based data logging can be used to store historical fault information for long-term analysis and performance evaluation. The system can be expanded to monitor multi-phase and high-voltage underground cables for wider industrial applications. Long-range communication modules such as LoRa or NB-IoT may be incorporated for extended coverage in large areas. Overall, the system can evolve into a comprehensive smart-grid monitoring solution for future electrical infrastructure.

REFERENCES

1. M. Sharma and R. Gupta, “*IoT-Based Monitoring System for Underground Cable Fault Detection,*” International Journal of Advanced Electrical Engineering, vol. 12, no. 3, pp. 145–152, 2021.
2. S. Khan, P. Verma, and A. Singh, “*Design and Implementation of an Embedded System for Underground Cable Fault Location,*” IEEE Transactions on Power Delivery, vol. 36, no. 4, pp. 987–995, 2020.
3. R. Patel and V. Mehta, “*Real-Time Cable Fault Identification Using Microcontroller and IoT Technology,*” Journal of Electrical Systems and Information Technology, vol. 8, no. 1, pp. 55–67, 2021.
4. A. N. Ansari et al., “*Smart Grid-Based Fault Monitoring Using Wireless Sensor Networks,*” International Journal of Electronics and Communication Engineering, vol. 14, no. 2, pp. 88–96, 2020.
5. T. S. Reddy and K. R. Kumar, “*Automated Fault Distance Estimation in Underground Power Cables Using Embedded Controllers,*” International Journal of Engineering Research and Technology, vol. 10, no. 6, pp. 210–216, 2022.
6. P. D. Deshmukh and S. B. Patil, “*A Novel IoT-Assisted Platform for Cable Fault Detection and Reporting,*” International Journal of Smart Sensor Technologies, vol. 7, no. 1, pp. 34–43, 2023.
7. M. Khan and S. Alam, “*Voltage Variation-Based Fault Detection Model for Subsurface Power Lines,*” International Conference on Electrical, Electronics and Automation, pp. 282–289, 2019.
8. R. K. Singh and D. Choudhary, “*Underground Cable Fault Analysis Using Resistive Divider and Microcontroller,*” Journal of Power Electronics and Drives, vol. 11, no. 2, pp. 112–120, 2022.
9. B. Joseph and L. Mathew, “*IoT-Integrated Smart Diagnostic System for Underground Electrical Networks,*” IEEE International Conference on Sustainable Energy Systems, pp. 399–406, 2021.